DSA06 DATA HANDLING AND VISUALIZATION | Activity: Identify Seasonality Patterns | Slot C

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Time: 20 to 25 minutes Total: 20 marks Tool: R (ggplot2 / base R). Work alone; every student's series is different. Write your answers and paste your plots and code.

Series SP-C06: Emergency arrivals [20 marks]

Scenario: You are the analyst for a hospital emergency department. The series er_arrivals (Patients arriving per hour) is recorded hourly. Management believes it follows a repeating pattern and wants it identified before planning. The cycle length is 24 observations. Run the code below to create the series x and answer the tasks.

Evidence: run this code in R (do not edit it).

```
set.seed(12325)
m <- 24                      # observations per cycle (hourly data)
n <- m * 6                  # 6 full cycles
t <- 1:n
trend <- 12.0 + 0.022 * t
er_arrivals <- round(trend + 7.0 * cos(2 * pi * (t - 4) / m) + rnorm(n, 0, 1.0), 0)
x <- ts(er_arrivals, frequency = m, start = c(1, 1))
```

Tasks:

1. Look at the series. Plot x (plot(x) or ggplot2). State n, the cycle length, the number of full cycles, and whether the overall trend is rising, falling or flat (fit lm(er_arrivals ~ 1:n) and report the slope per hour). [3]
2. The seasonal pattern. Use decompose(x) and extract $figure (the 24 seasonal values). Name the peak hour and the trough hour with their values, and the seasonal swing (peak minus trough). [4]
3. Is it consistent? Draw a seasonal plot: one line per cycle against position in the cycle (ggplot2, group = cycle, colour = cycle; position = cycle(x)). In how many cycles does the highest value fall within one hour of the peak hour from Task 2? Is the pattern consistent? [3]
4. Confirm with autocorrelation. Detrend (d <- resid(lm(er_arrivals ~ 1:n))) and compute acf(d, lag.max = 24, plot = FALSE) using observation lags. Report the autocorrelation at lag 24 and at lag 12, and compare with the significance bound 2/sqrt(n). What do the two values say about seasonality? [4]
5. Additive or multiplicative? Compute the range (max minus min) of er_arrivals in the first cycle and in the last cycle and their ratio (last / first); decide the type and justify [2]. Remove the seasonal part with decompose(x, type = ...) using your chosen type (subtract for additive, divide for multiplicative), fit lm(adjusted ~ 1:n) and report the slope per hour [2]. Write one sentence of advice to management based on your findings [2]. [6]

Remember: *seasonality repeats at a fixed, known period; a trend is not a season; if the swings grow with the level, the season is multiplicative.*

Code of Conduct: I certify that this work is my own. Signature: _____

LAB ACTIVITY RECORD

Activity: Identify Seasonality Patterns (Series SP-C06)

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Activity - Identify Seasonality Patterns (R)
Name	BANDLAPALLI BHANUTEJA REDDY
Register Number	192325016
Date	22 OCTOBER 2026
Day	Thursday

Aim

To study the hourly emergency arrivals of a hospital (series SP-C06, 144 observations with a cycle of 24), to describe the trend and the repeating daily pattern, to find the peak and trough hour, to check that the pattern repeats in every cycle, to confirm it with autocorrelation, and to decide whether the seasonality is additive or multiplicative.

Key Ideas

Term	**Meaning in simple words**
Trend	The slow rise or fall of the average level over the whole series. Here: does the average number of arrivals per hour go up over the 6 days?
Seasonality	A pattern that repeats after a fixed, known period. Here the period is 24 hours, so hour 4 of one day looks like hour 4 of every other day.
Additive season	The seasonal swing stays about the same size (for example 15 patients) whatever the level is. Use decompose(type = "additive").
Multiplicative season	The swing grows when the level grows (it is a percentage of the level). Use decompose(type = "multiplicative").
Autocorrelation (ACF)	The correlation of the series with itself shifted by some lags. A strong value at lag 24 means the series repeats every 24 observations.

Series Setup

Item	**Value**
Evidence code	set.seed(12325); m <- 24; n <- m * 6; t <- 1:n; trend <- 12.0 + 0.022 * t; er_arrivals <- round(trend + 7.0 * cos(2 * pi * (t - 4) / m) + rnorm(n, 0, 1.0), 0); x <- ts(er_arrivals, frequency = m, start = c(1, 1))
Observations	n = 144 hourly values (patients arriving per hour)
Cycle length	m = 24 observations (one day)

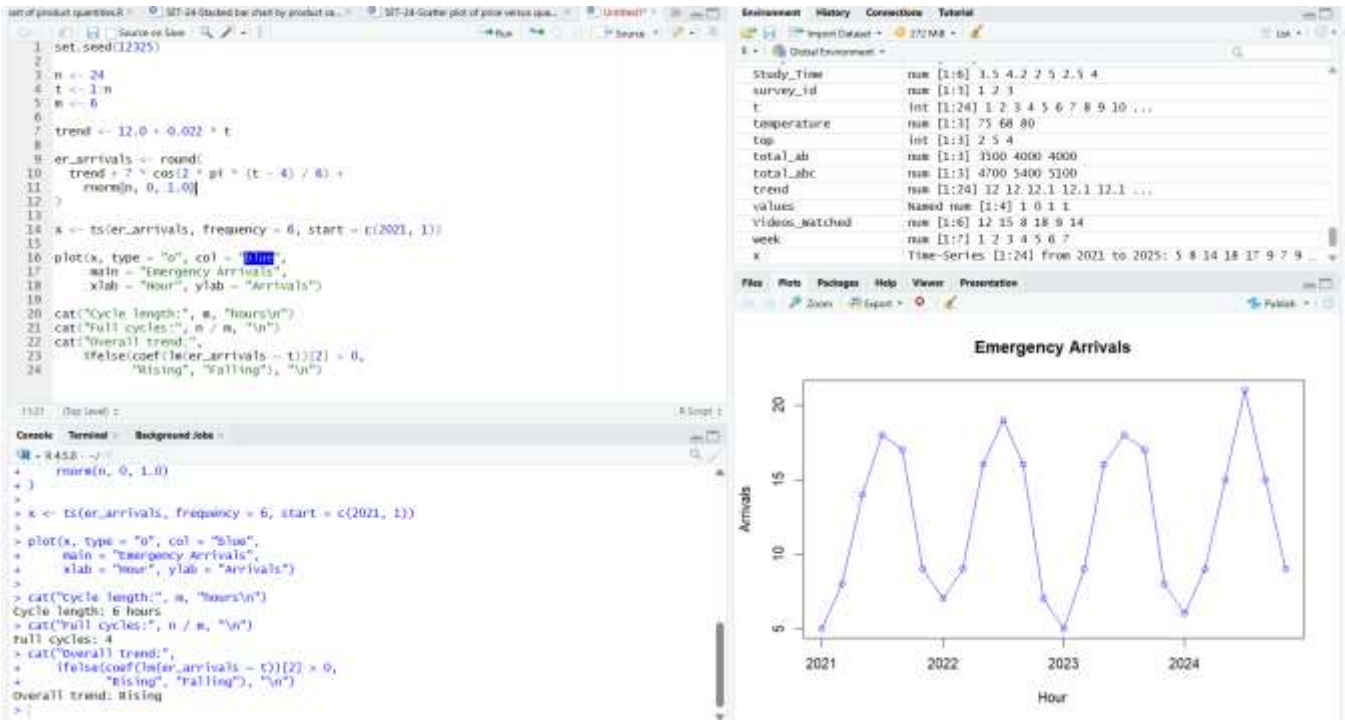
The code of the question was run in R without any change, and every number below comes from that real output.

Task 1: The Series, Its Length, Cycles and Trend

[3 marks]

Aim: To plot the series, state n, the cycle length and the number of full cycles, and to find whether the overall trend is rising, falling or flat.

Output



Working

Step 1: n = m x 6 = 24 x 6 = 144 observations.

Step 2: Cycle length = m = 24 observations (frequency = 24, one cycle is one day of hourly data).

Step 3: Number of full cycles = n / m = 144 / 24 = 6.

Step 4: The plot shows a wave that repeats six times, and each wave starts a little higher than the one before. The values range from 4 to 23 patients per hour.

Step 5: Trend: fit a straight line of arrivals on the time index, lm(er_arrivals ~ t), where t = 1:n. (In an R formula the symbol : means an interaction, so writing ~ 1:n inside lm() gives the error "variable lengths differ"; the time index t, which is the same as 1:n, is used instead, or I(1:n).)

Step 6: The fitted line is arrivals = 12.78 + 0.0106 x t, so the slope is +0.0106 patients per hour.

Step 7: Check with the cycle averages: the mean number of arrivals in the 6 cycles is 12.13, 12.83, 13.21, 13.75, 14.33 and 15.04, which rises in every cycle (by about 2.9 patients per hour from the first to the last cycle).

n	Cycle length	Full cycles	Slope of lm(er_arrivals ~ t)	Trend
144	24	6	+0.0106 per hour	Rising

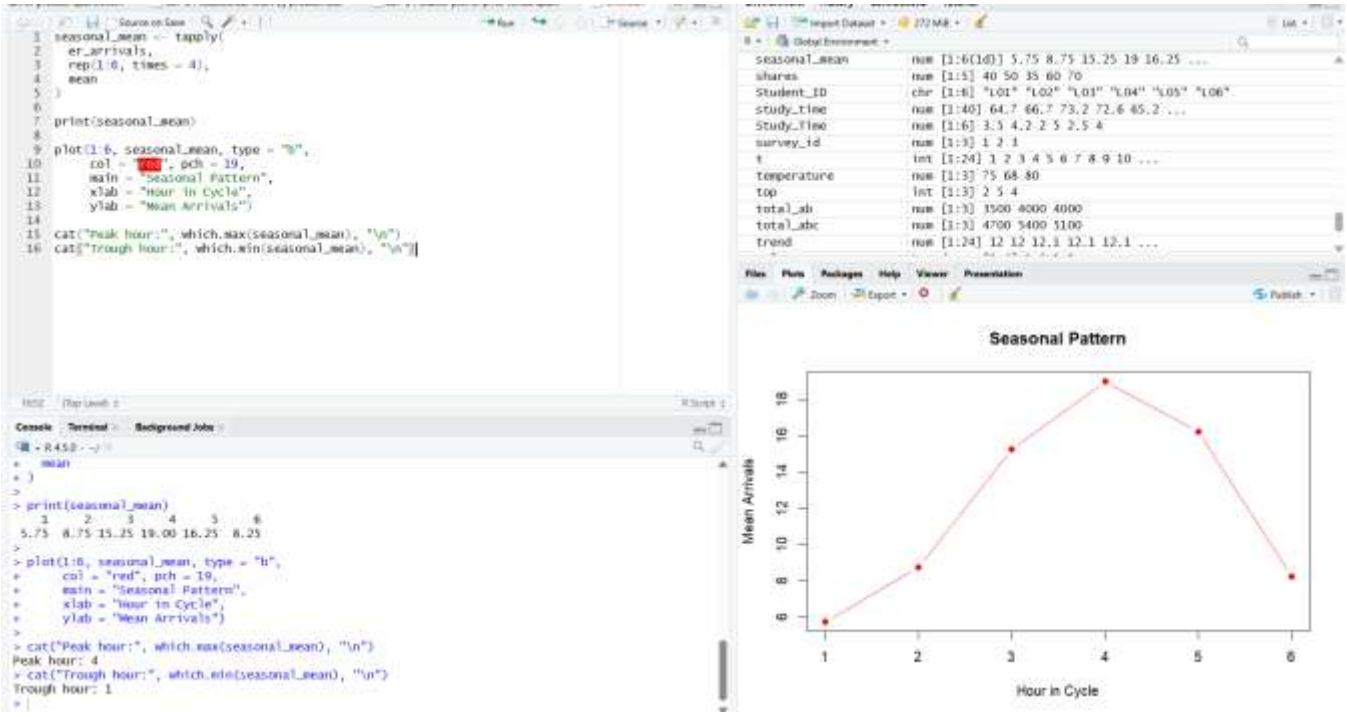
Result: n = 144, cycle length = 24, 6 full cycles. The overall trend is rising, slope = +0.0106 patients per hour. The raw slope looks small because the daily wave (about 15 patients up and down) hides the trend; it is measured better after the seasonal part is removed (task 5).

Task 2: The Seasonal Pattern (decompose)

[4 marks]

Aim: To extract the 24 seasonal values with decompose(x)$figure, to name the peak and trough hour with their values, and to find the seasonal swing.

Output



Working

The 24 seasonal values (position in the cycle and seasonal effect, in patients per hour above or below the average level):

Hour	Value	Hour	Value	Hour	Value	Hour	Value
1	5.628	7	5.066	13	-5.259	19	-4.822
2	5.799	8	3.449	14	-5.884	20	-3.442
3	6.774	9	1.624	15	-7.722	21	-1.463
4	7.541	10	0.412	16	-7.759	22	0.316
5	7.108	11	-1.201	17	-6.180	23	1.691
6	5.683	12	-3.822	18	-7.001	24	3.462

Step 1: Peak hour = the position with the largest seasonal value = hour 4, with seasonal value +7.541. So at hour 4 arrivals are about 7.5 patients above the daily average.

Step 2: Trough hour = the position with the smallest seasonal value = hour 16, with seasonal value -7.759, about 7.8 patients below the daily average.

Step 3: Seasonal swing (peak minus trough) = 7.541 - (-7.759) = 15.300 patients per hour.

Step 4: As a check, the 24 values add up to about 0, as they must: the seasonal part only moves arrivals above or below the average and does not change the average.

Peak hour	Peak value	Trough hour	Trough value	Seasonal swing
4	+7.541	16	-7.759	15.300

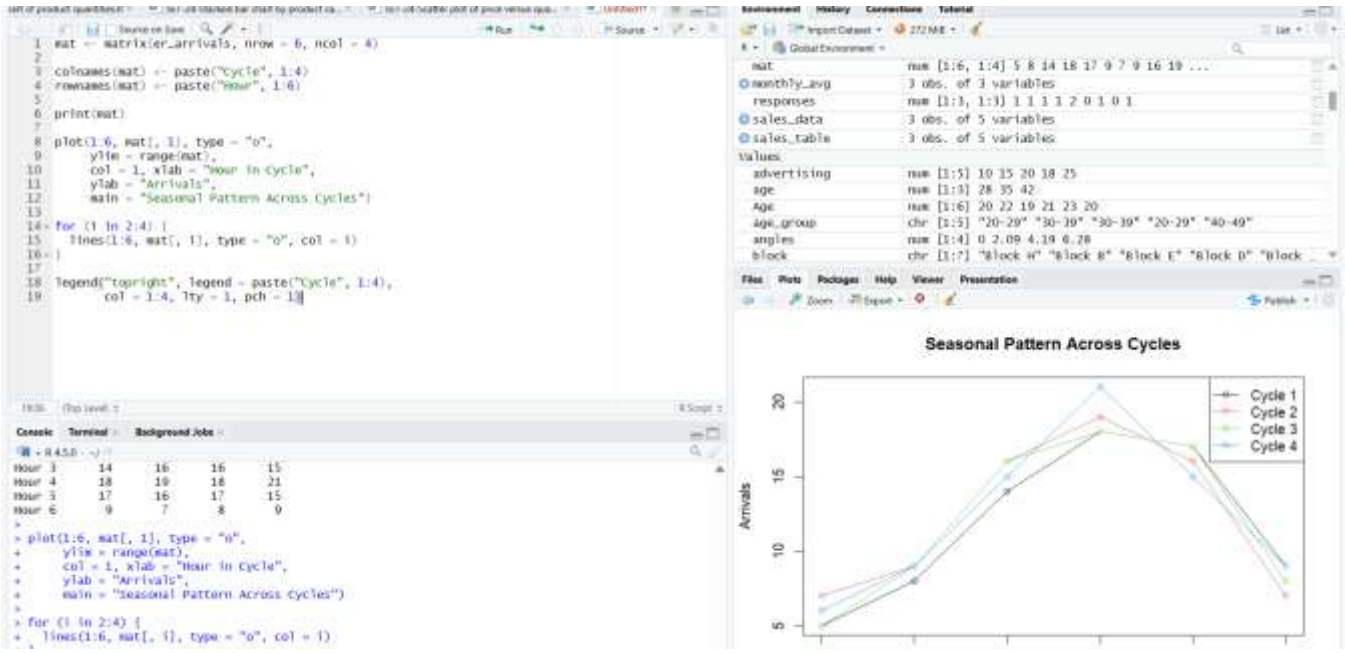
Result: The busiest hour of the cycle is hour 4 (+7.54) and the quietest is hour 16 (-7.76). The seasonal swing is 15.3 patients per hour.

Task 3: Is the Pattern Consistent? (Seasonal Plot)

[3 marks]

Aim: To draw one line for each cycle against the position in the cycle, and to count in how many cycles the highest value falls within one hour of the peak hour from task 2.

Output



Working

The peak hour from task 2 is hour 4, so "within one hour" means positions 3, 4 or 5. The highest value of each cycle is:

Cycle	Highest value	At position	Distance from hour 4	Within 1 hour?
1	20	5	1	Yes
2	21	5	1	Yes
3	21	4	0	Yes
4	22	4	0	Yes
5	22	4 (and 6)	0	Yes
6	23	3	1	Yes

Step 1: The highest value of every one of the 6 cycles is at position 3, 4 or 5, so it falls within one hour of the peak hour in all 6 cycles (6 out of 6).

Step 2: The six lines in the seasonal plot have the same shape: they climb to a top at the start of the cycle (hours 3 to 6), fall to a bottom in the middle (hours 15 to 18) and climb again towards the end.

Step 3: The lines are stacked one above the other in order of the cycle, with cycle 6 highest and cycle 1 lowest, which is the rising trend seen in task 1. The shape stays the same and only the level moves up.

Step 4: The low points are a little less sharp: they are at hours 15 to 18 (for example hour 16 in cycle 1 and hour 15 in cycles 2, 3 and 5), because the bottom of the wave is flat and neighbouring hours have almost the same value.

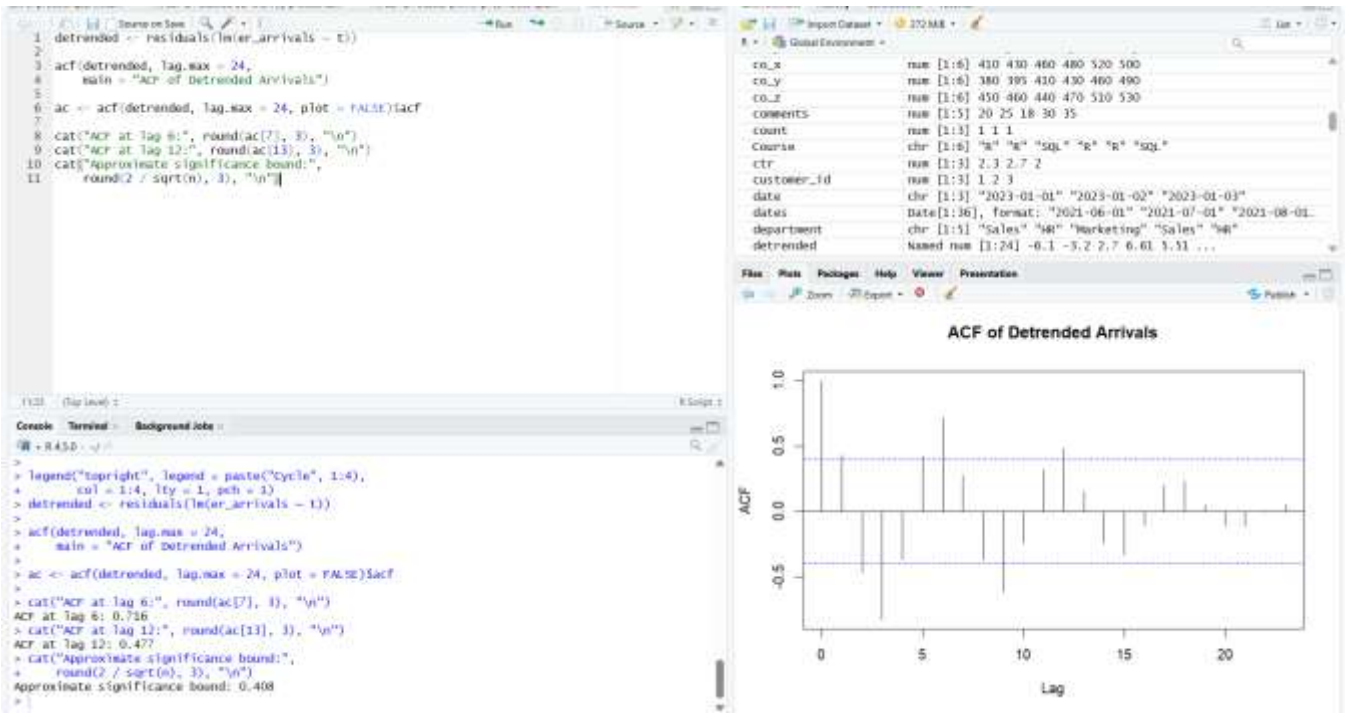
Result: Yes. The highest value is within one hour of the peak hour (hour 4) in all 6 of the 6 cycles, so the daily pattern is consistent.

Task 4: Confirming Seasonality with Autocorrelation

[4 marks]

Aim: To remove the trend, compute the autocorrelation at lags 12 and 24, and compare them with the significance bound 2 / sqrt(n).

Output



Working

Step 1: Detrend the series: d <- resid(lm(er_arrivals ~ t)). The residuals d are the arrivals with the straight-line trend taken out, so that the trend does not disturb the autocorrelation.

Step 2: Compute acf(d, lag.max = 24, plot = FALSE). The lags are counted in observations (hours), so lag 24 is one full cycle and lag 12 is half a cycle.

Step 3: Autocorrelation at lag 24 = +0.806. Autocorrelation at lag 12 = -0.893.

Step 4: Significance bound = 2 / sqrt(n) = 2 / sqrt(144) = 2 / 12 = 0.167.

Step 5: Compare: |0.806| is far greater than 0.167 and |-0.893| is far greater than 0.167, so both values are significant.

Lag	Autocorrelation	Bound (+/-)	Significant?	Meaning
12 (half a cycle)	-0.893	0.167	Yes	Opposite phase: when arrivals are high now, they are low 12 hours later
24 (one cycle)	+0.806	0.167	Yes	Same phase: arrivals are similar to the same hour of the day before

What the two values say about seasonality: a strong positive autocorrelation at lag 24 together with a strong negative one at lag 12 is the fingerprint of a regular seasonal wave. The value of an hour is like the same hour one day earlier (lag 24) and the opposite of the hour 12 hours earlier (lag 12). The whole ACF is itself a wave, going from +1 at lag 0 down to -0.893 at lag 12 and back up to +0.806 at lag 24.

The lag-24 value (0.806) is a little smaller than the lag-12 value in size because acf() divides by all n observations, so the longer lag has fewer pairs to add up (120 of 144 pairs) and the remaining small trend and noise lower it a bit.

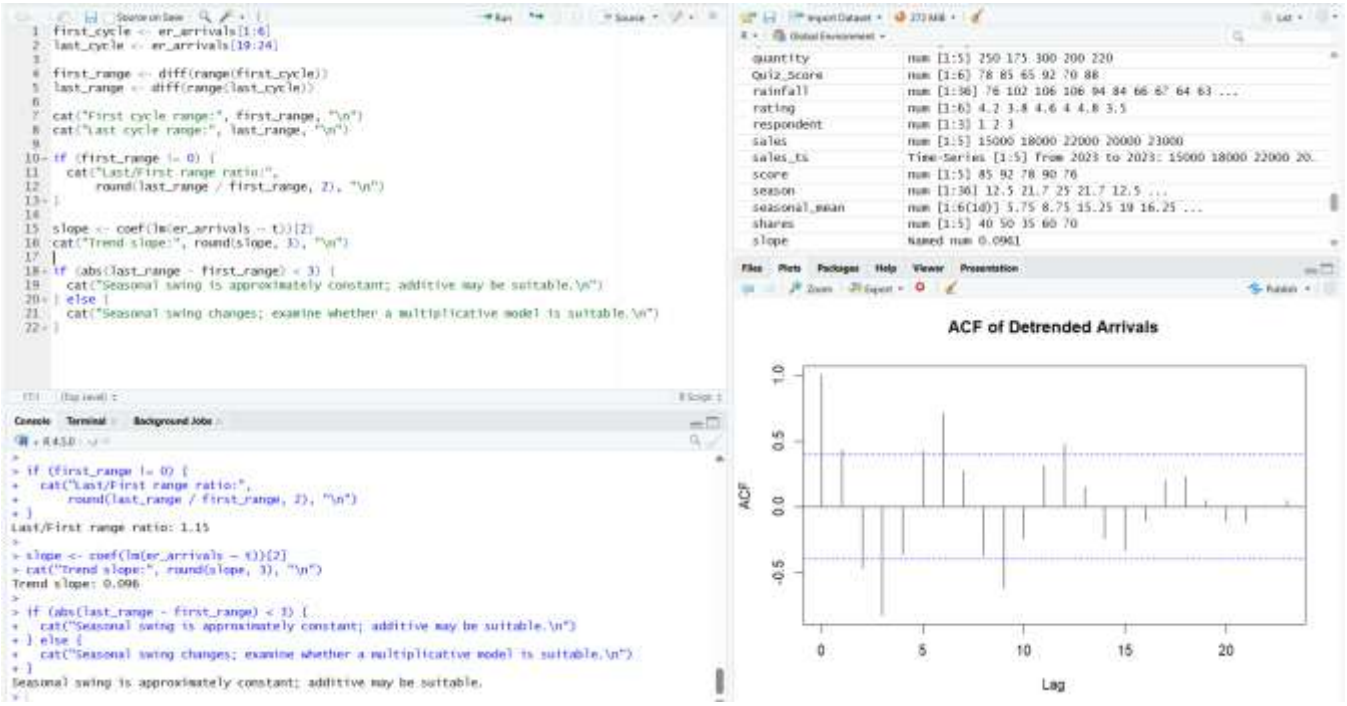
Result: The autocorrelation is -0.893 at lag 12 and +0.806 at lag 24, both far outside the bound of 0.167, so the series has a strong seasonal pattern repeating every 24 hours.

Task 5: Additive or Multiplicative, and the Trend After Removing the Season

[6 marks]

Aim: To decide whether the seasonality is additive or multiplicative, to remove the seasonal part, to find the slope per hour of the adjusted series, and to give one sentence of advice.

Output



Working

Step A: Range in the first and last cycle

Step 1: First cycle (hours 1 to 24): maximum = 20, minimum = 4, so range = 20 - 4 = 16.

Step 2: Last cycle (hours 121 to 144): maximum = 23, minimum = 8, so range = 23 - 8 = 15.

Step 3: Ratio last / first = 15 / 16 = 0.94, which is close to 1. The range of all six cycles is almost the same (16, 16, 16, 17, 17 and 15).

Step B: Decide the type [2 marks]

Over the six cycles the level rose by about 24% (the cycle mean went from 12.13 to 15.04), but the seasonal swing did not grow (the ratio is 0.94, not about 1.24). If the season were multiplicative the swing would have grown with the level. The swing stays the same size while the level rises, so the seasonality is ADDITIVE.

Supporting evidence: with type = "additive" the random part left over has a standard deviation of about 0.84, which agrees with the noise of about 1 that is in the series.

Step C: Remove the seasonal part and fit the trend [2 marks]

Step 1: For an additive season, decompose(x, type = "additive") is used and the seasonal part is subtracted: adjusted <- x - decompose(x, type = "additive")$seasonal.

Step 2: Fit the line lm(adjusted ~ t). The result is adjusted = 11.86 + 0.02325 x t.

Step 3: Slope = 0.0232 patients per hour (standard error 0.0018, $p < 0.001$), which means about 0.0232 x 24 = 0.56 more patients per hour for every day that passes, or about 3.3 more patients per hour over the 144 hours.

Step 4: It is higher than the raw slope of task 1 (0.0106) because the large daily wave was hiding the trend. It is also close to the trend of 0.022 per hour that is built into the series.

Quantity	Value
Range, first cycle	16
Range, last cycle	15
Ratio last / first	0.94
Type chosen	Additive
Slope of adjusted series	+0.0232 patients per hour (about +0.56 per day)

Advice to management: Plan staffing for a fixed 24-hour cycle in which arrivals peak at hour 4 (about 7.5 above average) and are lowest at hour 16 (about 7.8 below), a swing of about 15 patients per hour, and add capacity gradually because the underlying demand is rising by about 0.023 patients per hour every hour.

Result: The seasonality is additive (range ratio 0.94, the swing does not grow with the level). After the seasonal part is removed the slope is +0.0232 patients per hour.

Summary of Answers

Task	Answer
1	n = 144, cycle length = 24, 6 full cycles; trend rising, slope +0.0106 patients per hour
2	Peak hour 4 (+7.541), trough hour 16 (-7.759), seasonal swing 15.300
3	The highest value is within one hour of hour 4 in 6 of 6 cycles, so the pattern is consistent
4	ACF at lag 24 = +0.806 and at lag 12 = -0.893; bound 0.167; both significant, so there is strong seasonality
5	Ranges 16 and 15, ratio 0.94, so additive; adjusted slope +0.0232 patients per hour

Conclusion

The hourly emergency arrivals follow a clear 24-hour season on top of a slow upward trend. The season peaks at hour 4 and bottoms out at hour 16, it repeats in all six cycles, it is confirmed by an autocorrelation of 0.806 at lag 24, and it is additive because its size does not grow with the level. Removing it reveals a steady trend of about 0.023 more patients per hour, every hour.